Special Numbers

Year 4 Pt 2 · **Lesson Worksheet**

Worked Examples

Square numbers & roots: Is 81 a square number? If so, find its square root.

Answer: Yes: 9 × 9 = 81, so 81 is a square number and √81 = 9.

Triangular numbers: Find the first five triangular numbers and describe the pattern used to generate them.

Answer: 1, 3, 6, 10, 15 — each triangular number is the sum of consecutive whole numbers starting at 1 (e.g. the 4th is 1+2+3+4 = 10), so each term adds one more than the last addition.

Cube numbers & sum of odd numbers: Show that the sum of the first 4 odd numbers (1+3+5+7) equals a square number.

Answer: 1 + 3 + 5 + 7 = 16 = 4^2, confirming that the sum of the first n odd numbers always equals n^2.

Square & Cube Basics (15 questions)

1. Write down the first six square numbers.
2. What is 7^2?
3. What is the square root of 49?
4. What is 2^3 (2 cubed)? A) 4 B) 6 C) 8 D) 10 E) 12
5. What is the cube root of 27?
6. Is 36 a square number? Explain how you know.
7. Write down the first four triangular numbers.
8. What is the 5th triangular number? A) 10 B) 12 C) 15 D) 18 E) 20
9. What is 10^2 ?
10. Which of these is a cube number? A) 16 B) 25 C) 36 D) 64 E) 90
11. Select ALL that apply: which of these numbers are square numbers? A) 9 B) 16 C) 20 D) 25 E) 32
12. What is the square root of 100?
13. What is 4^3 (4 cubed)?
14. Select ALL that apply: which of these numbers are triangular numbers? A) 6 B) 8 C) 10 D) 14 E) 15
15. What is $3^2 + 2^3$? A) 15 B) 16 C) 17 D) 18 E) 19

Patterns & Classifying (15 questions)

1. A number is both a square number and an even number, and it lies between 50 and 100. What is the number?
2. The sum of the first 3 odd numbers is 1 + 3 + 5 = 9. Use this pattern to write the sum of the first 6 odd numbers without adding them one by one.
3. What is the 7th triangular number?

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4. Which of these numbers is a triangular number? A) 21 B) 22 C) 23 D) 24 E) 25

5. Select ALL that apply: which of these statements about 64 are true? A) 64 is a square number. B) 64 is a cube number. C) The square root of 64 is 8. D) The cube root of 64 is 4. E) 64 is an odd number.

6. Find two triangular numbers that add together to make 25.

7. What is the difference between the 6th and 5th square numbers?

8. A number is a cube number greater than 30 and less than 100. What is the number? A) 27 B) 36 C) 49 D) 64 E) 81

9. The 8th triangular number is 36. What is the 9th triangular number? (Hint: each new term adds one more than the previous term added.)

10. Select ALL that apply: which of these numbers are both square numbers and triangular numbers? A) 1 B) 9 C) 16 D) 36 E) 49

11. What is the sum of the first 5 odd numbers? What type of special number does this give?

12. Which square number, when you find its square root, gives an odd number greater than 8? A) 36 B) 49 C) 64 D) 81 E) 100

13. A gardener arranges tins of paint in a triangular stack: 1 tin in the top row, 2 in the next, 3 in the next, and so on. How many tins are in a stack with 6 rows?

14. What is $5^2 - 3^2$? A) 14 B) 15 C) 16 D) 17 E) 18

15. Explain, using an example, why a triangular number plus the next triangular number always gives a square number.

Special Numbers Challenge (15 questions)

1. Find the smallest square number that is greater than 150.

2. Find the largest cube number that is less than 200.

3. A square number has a square root that is itself a square number greater than 1. Find the smallest such square number. A) 4 B) 9 C) 16 D) 25 E) 36

4. The 10th triangular number is 55. Use this to find the 11th triangular number without listing all the terms.

5. Select ALL that apply: a number n is both a triangular number and an even number, with $10 < n < 50$. Which of these values of n are correct? A) 15 B) 21 C) 28 D) 36 E) 45

6. Show that $7^2 + 7$ equals twice the 7th triangular number. (The nth triangular number is $n(n+1)/2$.)

7. A square garden has an area of 169 m^2. What is the length of one side of the garden?

8. Find a square number from the list below that is also one more than a multiple of 10. A) 25 B) 36 C) 49 D) 64 E) 81

9. The 6th triangular number is 21. Find the 7th and 8th triangular numbers.

10. A cube number is formed by cubing a whole number n. If this cube number is 729, what is n? A) 7 B) 8 C) 9 D) 10 E) 11

11. Select ALL that apply: which of these statements about square and cube numbers are true? A) Every cube number is also a square number. B) 1 is both a square number and a cube number. C) 64 is both a square number and a cube number. D) All square numbers are even. E) For any whole number n, the nth triangular number plus the $(n-1)$th triangular number equals n^2.

12. A stack of oranges is built in triangular layers like the triangular number pattern: 1 orange on top, then 3, then 6, and so on. How many oranges are needed to build a stack 8 layers deep?

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13. Find the value of n if $n^2 = 225$. A) 13 B) 14 C) 15 D) 16 E) 17

14. A number is both a triangular number and a square number, and it is greater than 1 and less than 50. What is the number?

15. Using the fact that the sum of the first n odd numbers equals n^2, find the sum of the first 10 odd numbers without adding them individually.

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